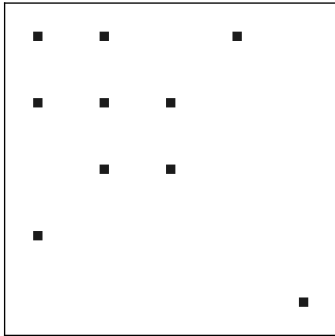


# MNA sparsity patterns across representative SMPS

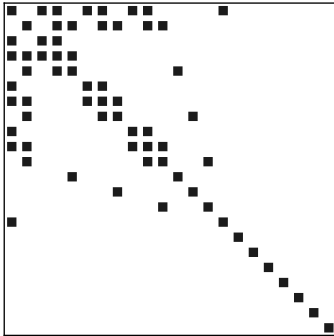
Buck (1 switch)

$n = 5$ , nnz = 10, density = 40.0%



NPC 3-phase (12 switches)

$n = 22$ , nnz = 66, density = 13.6%



MMC arm, 9 SMs (18 switches)

$n = 30$ , nnz = 66, density = 7.3%

